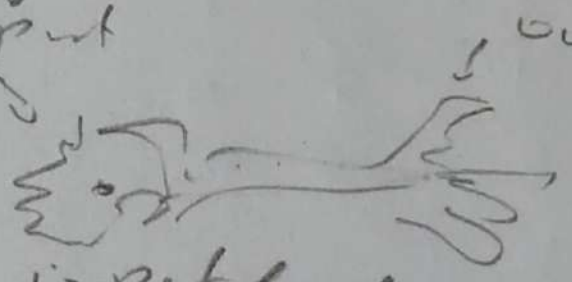
Introduction to machine learning -6-

Neural networks

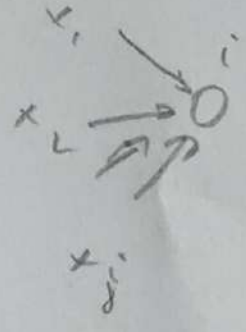
Tensorflow playground example
- circle dataset, relU
- no hidden layer → SVM
  ↳ x_1^2, x_2^2 kernels
- one hidden layer and only x_1, x_2
  ↳ 2, 3 neurons
- show linearity

Perceptron



- neuron
- threshold on input level

Representation:



activation function

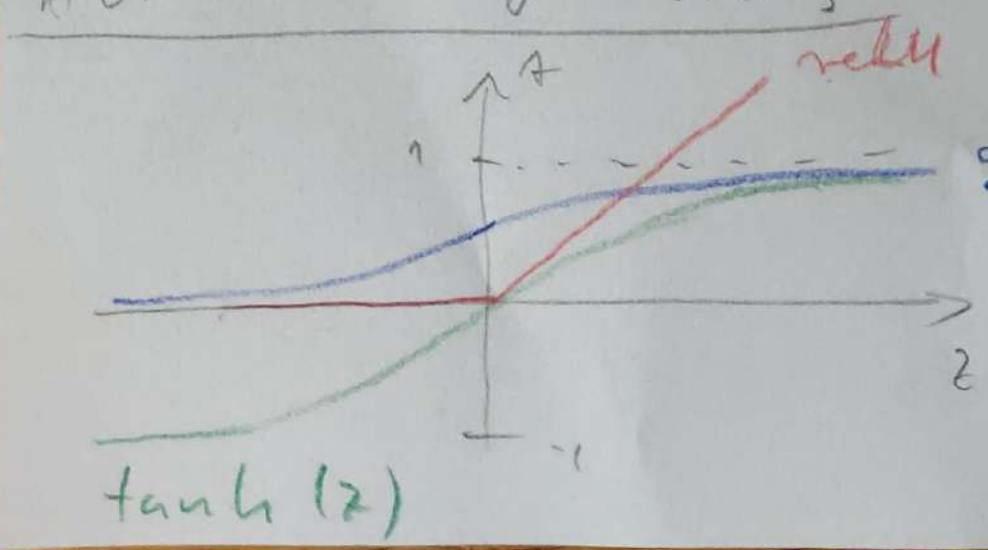
$$y = \phi(\underline{w}^T \underline{x} + b)$$

General notation:

$$z_i = \sum_j W_{ij} A_j + b_i \qquad A_i = g(z_i)$$

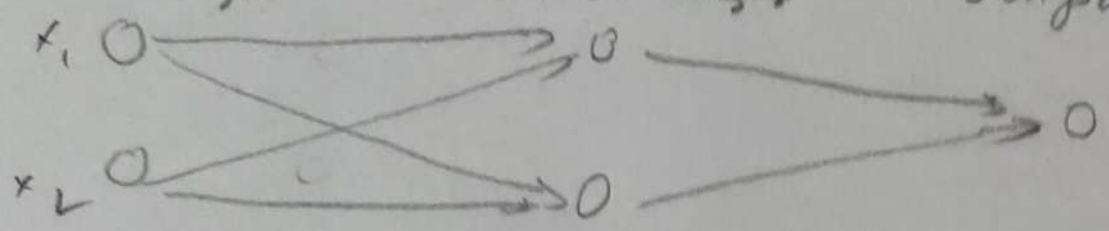
activation function

Activation functions



Let us try the XOR function

x_1	x_2	XOR
0	0	0
1	0	1
0	1	1
1	1	0



$\underline{b} = 0$

ReLU

$\underline{\underline{W}}^{(1)}$ $\underline{\underline{W}}^{(2)}$

$\underline{A}^{(0)} = \underline{x}$ $\underline{h} = \underline{A}^{(1)} = g(\underline{\underline{W}}^{(1)} \underline{A}^{(0)})$ $\underline{y} = \underline{A}^{(2)} = g(\underline{\underline{W}}^{(2)} \underline{A}^{(1)})$

Solution: e.g. $\underline{\underline{W}}^{(1)} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

x_1	x_2	$Z_1^{(1)}$	$Z_2^{(1)}$	$A_1^{(1)}$	$A_2^{(1)}$
0	0	0	0	0	0
0	1	−1	1	0	1
1	0	1	−1	1	0
1	1	0	0	0	0

$\underline{\underline{W}}^{(2)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

x_1	x_2	$A_1^{(1)}$	$A_2^{(1)}$	$A_1^{(2)}$
0	0	0	0	0
0	1	0	1	1
1	0	1	0	1
1	1	0	0	0

✓

Back propagation IML - 6 -

Training set: x_i, t_i, $Loss = \frac{1}{2}(y_i - t_i)^2$

y_i is the output for input vector x_i

? How to change the weights to decrease the error?

$\Delta W = -\alpha \frac{\partial L}{\partial W}$ α learning rate

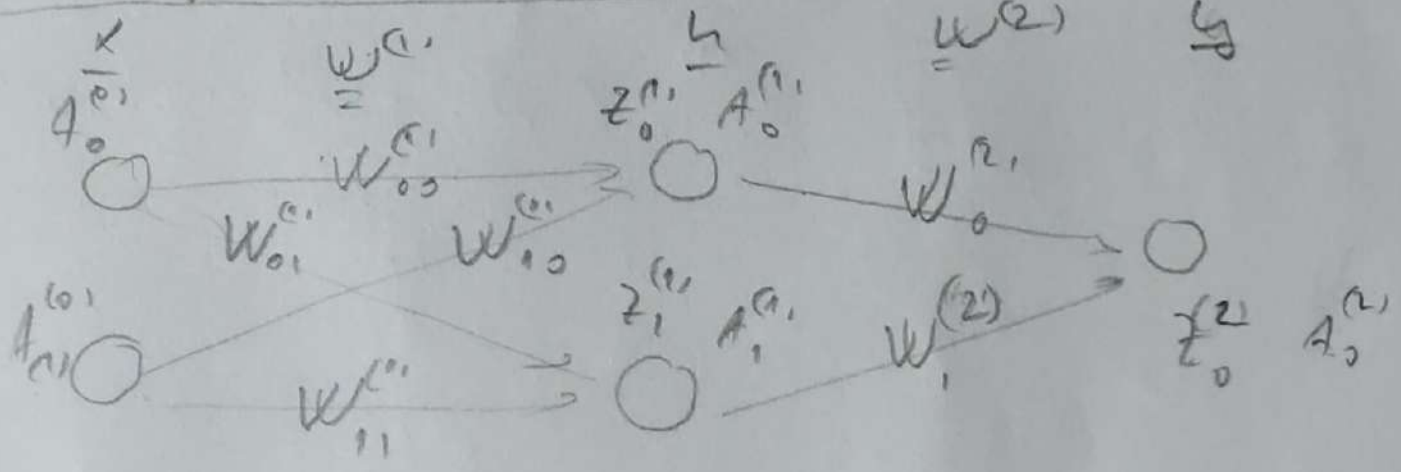
$\frac{\partial L}{\partial W} = ?$ → Solution Geoffrey Hinton (1986)

chain derivation rule!

$$\frac{\partial L}{\partial w_{ij}} = \frac{\partial L}{\partial A_i} \frac{\partial A_i}{\partial z_i} \frac{\partial z_i}{\partial w_{ij}} = \frac{\partial L}{\partial A_i} g'(z_i) A_j$$

$$z_i = \sum_j w_{ij} A_j + b_i$$

Example back propagation:



$\Delta w_0^{(2)} = ?$

$$\frac{\partial \mathcal{L}}{\partial w_0^{(2)}} = \frac{\partial \mathcal{L}}{\partial A_0^{(2)}} \frac{\partial A_0^{(2)}}{\partial w_0^{(2)}}$$

$$\mathcal{L} = \frac{1}{2}\left(A_0^{(2)} - t\right)^2 \qquad \frac{\partial \mathcal{L}}{\partial A_0^{(2)}} = \left(A_0^{(2)} - t\right) = \Delta$$

$$A_0^{(2)} = g\left(A_0^{(1)} w_0^{(2)} + A_1^{(1)} w_1^{(2)}\right) \qquad \frac{\partial A_0^{(2)}}{\partial w_0^{(2)}} = g'(z_0^{(2)}) \cdot \frac{\partial z_0^{(2)}}{\partial w_0^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial w_0^{(2)}} = \Delta \cdot g'\left(z_0^{(2)}\right) A_0^{(1)}$$

Similarly

$$\frac{\partial \mathcal{L}}{\partial w_1^{(2)}} = \Delta \cdot g'\left(z_1^{(2)}\right) A_1^{(1)}$$

Next layer

$$\frac{\partial \mathcal{L}}{\partial w_{00}^{(1)}} = \frac{\partial \mathcal{L}}{\partial A_0^{(2)}} \frac{\partial g\left(A_0^{(1)} w_0^{(2)} + A_1^{(1)} w_1^{(2)}\right)}{\partial w_{00}^{(1)}}$$

$w_{00}^{(1)}$ is only in this term

$$\frac{\partial \mathcal{L}}{\partial w_{00}^{(1)}} = \Delta\, g'(z_0^{(2)}) \frac{\partial \left[g\left(x_0 w_{00}^{(1)} + x_1 w_{10}^{(1)}\right) w_0^{(2)} + z_1^{(1)} w_1^{(2)}\right]}{\partial w_{00}^{(1)}}$$

$$\frac{\partial \mathcal{L}}{\partial w_{00}^{(1)}} = \Delta\, g'(z_0^{(2)})\, g'(z_0^{(1)})\, w_0^{(2)}\, A_0^{(0)}$$

$\| \; x_0$

Summary

$$\frac{\partial L}{\partial w_i^{(2)}} = \Delta \cdot g'(z_i^{(2)}) A_i^{(1)}$$

$$\frac{\partial L}{\partial w_{ij}^{(1)}} = \Delta \cdot g'(z_0^{(2)}) g'(z_j^{(1)}) w_j^{(2)} A_i^{(0)} = \Delta \cdot g'(z_0^{(2)}) g'(z_j^{(1)}) w_j^{(2)} x_i$$

Bias: we do not set the last multiplier with $A_i^{(1)}$ so

$$\frac{\partial L}{\partial b_i^{(2)}} = \Delta \cdot g'(z_i^{(2)})$$

$$\frac{\partial L}{\partial b_j^{(1)}} = \Delta \cdot g'(z_0^{(2)}) g'(z_j^{(1)}) w_j^{(2)}$$

Cost:

- backward pass is about twice as expensive as a forward pass
- cost is linear in number of layers and quadratic in number of units per layer